

Variable-Coefficient Order-Zero Operators on Compact Fractafolds Are Calderón–Zygmund

Run fa_banach_001 — candidate full solution

August 11, 2026

Abstract

Ionescu, Rogers, and Strichartz conjectured that their variable-coefficient order-zero pseudodifferential operators on compact fractafolds are Calderón–Zygmund. We prove the conjecture in the standing compact, boundaryless setting of their Section 9. Their eigenfunction expansion reduces the kernel to an absolutely summable family of constant-coefficient kernels. They summed only the size estimates; the standard-kernel difference estimates are uniform over the same family and sum identically. The obstruction that smooth functions need not be closed under multiplication is therefore bypassed.

1 The source conjecture

Let X be a compact fractafold without boundary in the setting of Section 9 of Ionescu–Rogers–Strichartz [1]. Write μ for its measure, R for the resistance metric, d for its resistance-metric dimension, and $-\Delta$ for its nonnegative Laplacian. A variable symbol $p \in S^0$ satisfies, for all integers $j, r \geq 0$,

$$\sup_{x \in X, \lambda > 0} \left| (\lambda \partial_\lambda)^r \Delta_x^j p(x, \lambda) \right| < \infty. \quad (1)$$

The associated operator is

$$Tu(x) = p(x, -\Delta)u(x) = \sum_{\lambda \in \text{spec}(-\Delta)} p(x, \lambda) P_\lambda u(x), \quad u \in \mathcal{D}.$$

The source proves that T is bounded on $L^2(\mu)$, has a continuous off-diagonal kernel satisfying $|K(x, y)| \lesssim R(x, y)^{-d}$, and is bounded on every L^q , $1 < q < \infty$. It then conjectures that T is a Calderón–Zygmund operator in the sense of [2].

In that cited definition, besides L^2 boundedness and the size estimate, a kernel is standard if, for some $c > 1$,

$$|K(x, y) - K(x, \bar{y})| \lesssim \frac{R(y, \bar{y})}{R(x, \bar{y})} R(x, y)^{-d} \quad (2)$$

whenever $R(x, y) \geq cR(y, \bar{y})$. The displayed modulus is Dini.

Let $n \geq 0$ be such that the series $S := \sum_k \lambda_k^{\alpha-n}$ converges. For $u \in \mathcal{D}$ we can write then

$$\begin{aligned} Tu(x) &= \sum_j \sum_k m_k(\lambda_j) \phi_k(x) P_{\lambda_j} u(x) \\ &= \sum_k \frac{\phi_k(x)}{\lambda_k^n} \sum_j \tilde{m}_{k,n}(\lambda_j) P_{\lambda_j} u(x) \\ &= \sum_k \frac{\phi_k(x)}{\lambda_k^n} \tilde{m}_{k,n}(-\Delta) u(x), \end{aligned}$$

where $\tilde{m}_{k,n}(-\Delta)$ is the pseudo-differential operator of order 0 attached to $\tilde{m}_{k,n}$. Theorem 3.6 and estimates (9.4) imply that $\tilde{m}_{k,n}(-\Delta)$ is given by integration with respect to a kernel $K_{k,n}(x, y)$ that is continuous (and smooth) off diagonal and satisfies the estimate

$$(9.6) \quad |K_{k,n}(x, y)| \leq C \cdot R(x, y)^{-d},$$

where C is a positive constant that is independent of k . Then

$$Tu(x) = \int \sum_k \frac{\phi_k(x)}{\lambda_k^n} K_{k,n}(x, y) u(y) d\mu(y).$$

Thus it suffices to prove that

$$K(x, y) = \sum_k \frac{\phi_k(x)}{\lambda_k^n} K_{k,n}(x, y)$$

is well defined and continuous. Estimates (9.6) imply that

$$|K(x, y)| \leq C \cdot R(x, y)^{-d} \sum_k \frac{\lambda_k^\alpha}{\lambda_k^n} = CS \cdot R(x, y)^{-d}.$$

Thus $K(x, y)$ is continuous off diagonal and (9.2) holds. $\square$

We conjecture that the kernel K of a pseudo-differential operator $p(x, -\Delta)$ of order 0 is a Calderón-Zygmund operator in the sense of [31, Section I.6.5]. Even though we are unable at this point to prove this claim, we can, nevertheless, show that these operators are bounded on $L^q(\mu)$, for all $1 < q < \infty$.

Theorem 9.6. *Suppose that X is a compact fractafold with no boundary and $p \in S^0$. Then the operator $Tu(x) = p(x, -\Delta)u(x)$ extends to a bounded operator on $L^q(\mu)$ for all $1 < q < \infty$.*

Proof. Fix $1 < q < \infty$ and let r such that $1/q + 1/r = 1$. Let $u \in \mathcal{D}$ and let $\tilde{m}_{k,n}$ as in the proof of Theorem 9.5. Recall that $\tilde{m}_{k,n}$ is a symbol of order 0 with constant coefficients and

Figure 1: The source's uniform kernel expansion and conjecture (arXiv:1108.2246, PDF page 27).

2 Idea of the proof

The source expands p in an eigenbasis in its x -variable. After multiplying the k th coefficient by a high power of its eigenvalue, one obtains a bounded family of constant-coefficient S^0 symbols. The associated kernels have uniform Calderón–Zygmund estimates. Dividing back introduces scalar coefficients whose sup norms are absolutely summable for a sufficiently high power.

The source uses this decomposition only to sum kernel sizes. But the Calderón–Zygmund difference is in the y -variable, while each expansion coefficient depends only on x . Thus the difference estimates sum by the same triangle inequality. No product of two functions is ever tested for membership in $\text{dom } \Delta$.

3 Uniform constant-coefficient pieces

Choose an orthonormal eigenbasis (ϕ_k) ,

$$-\Delta\phi_k = \lambda_k\phi_k, \quad \lambda_k \geq 0,$$

and set

$$m_k(\lambda) = \int_X p(z, \lambda) \overline{\phi_k(z)} d\mu(z).$$

We use $1 + \lambda_k$ rather than λ_k in order to include a possible zero mode without a separate case.

Lemma 1. *For every integer $N \geq 0$, the symbols*

$$b_{k,N}(\lambda) = (1 + \lambda_k)^N m_k(\lambda)$$

form a bounded family in S^0 : every S^0 seminorm is bounded independently of k .

Proof. Self-adjointness and the absence of boundary give

$$b_{k,N}(\lambda) = \int_X (I - \Delta_z)^N p(z, \lambda) \overline{\phi_k(z)} d\mu(z). \quad (3)$$

The operators $\lambda\partial_\lambda$ and Δ_z commute. Hence, for every $r \geq 0$, Cauchy–Schwarz and $\|\phi_k\|_2 = 1$ yield

$$|(\lambda\partial_\lambda)^r b_{k,N}(\lambda)| \leq \mu(X)^{1/2} \max_{z \in X} |(I - \Delta_z)^N (\lambda\partial_\lambda)^r p(z, \lambda)|.$$

Expanding $(I - \Delta_z)^N$ and using (1) gives a bound depending on N, r, p , but not on k or λ . $\square$

Let $B_{k,N} = b_{k,N}(-\Delta)$ and let $H_{k,N}(x, y)$ denote its kernel. The constant-coefficient kernel theorem of [1, Theorem 3.6], together with the standard-kernel criterion of [2], gives constants independent of k such that

$$|H_{k,N}(x, y)| \leq C_N R(x, y)^{-d}, \quad (4)$$

$$|H_{k,N}(x, y) - H_{k,N}(x, \bar{y})| \leq C_N \frac{R(y, \bar{y})}{R(x, \bar{y})} R(x, y)^{-d} \quad (5)$$

under the separation condition in (2). Uniformity follows from Lemma 1: the proof of the constant-coefficient theorem uses finitely many symbol seminorms for each displayed kernel estimate. In particular it first proves, uniformly,

$$|\Delta_y H_{k,N}(x, y)| \leq C_N R(x, y)^{-2d-1},$$

from which (5) follows by the Ionescu–Rogers criterion.

4 Summing the standard kernels

Theorem 1. *Under the standing compact, boundaryless hypotheses of Section 9 of [1], every variable-coefficient order-zero operator $p(x, -\Delta)$ is a Calderón–Zygmund operator in the precise sense cited by the source. Consequently it is of weak type $(1, 1)$ and bounded on $L^q(\mu)$ for every $1 < q < \infty$.*

Proof. The eigenfunction estimate and compact spectral summability used in the proof of [1, Theorem 9.5] allow us to choose N so large that

$$A_N := \sum_k \frac{\|\phi_k\|_\infty}{(1 + \lambda_k)^N} < \infty. \quad (6)$$

Indeed, the source proves $\|\phi_k\|_\infty \leq C(1 + \lambda_k)^\alpha$ for some α and then chooses an arbitrarily high spectral power with a summable tail.

The eigenfunction expansion of $p(\cdot, \lambda)$ gives, first on finite spectral sums,

$$\begin{aligned} Tu(x) &= \sum_k \phi_k(x) m_k(-\Delta) u(x) \\ &= \sum_k \frac{\phi_k(x)}{(1 + \lambda_k)^N} B_{k,N} u(x). \end{aligned}$$

Thus the off-diagonal kernel found in the source is

$$K(x, y) = \sum_k \frac{\phi_k(x)}{(1 + \lambda_k)^N} H_{k,N}(x, y). \quad (7)$$

The series is locally uniformly convergent off the diagonal by (6) and (4). Moreover,

$$|K(x, y)| \leq C_N A_N R(x, y)^{-d}.$$

For $R(x, y) \geq cR(y, \bar{y})$, subtract (7) at y and $\bar{y}$, then use (5):

$$\begin{aligned} |K(x, y) - K(x, \bar{y})| &\leq \sum_k \frac{|\phi_k(x)|}{(1 + \lambda_k)^N} |H_{k,N}(x, y) - H_{k,N}(x, \bar{y})| \\ &\leq C_N A_N \frac{R(y, \bar{y})}{R(x, \bar{y})} R(x, y)^{-d}. \end{aligned}$$

This is (2) with a linear Dini modulus. The source already proved the required L^2 boundedness and kernel representation. Hence T is Calderón–Zygmund. The endpoint and L^q conclusions are the standard consequences recorded in [2]; the strong L^q part was also proved directly in the source. $\square$

5 Verification, scope, and novelty

There is a second route to the same result: the estimates above permit termwise application of Δ_y in (7); the resulting series converges locally uniformly off the diagonal and is bounded by $CA_N R(x, y)^{-2d-1}$. Applying the Ionescu–Rogers theorem directly then recovers (2).

The proof does not claim an extension to infinite blow-ups. In that setting the compact eigenbasis and summability step (6) need a replacement. It also uses the one-sided-in- y standard-kernel definition explicitly cited by the source; no assertion of smoothness in the x variable is made.

A bounded novelty search covered the run’s registry, solution, attempt, and proof-gap indexes; exact title and conjecture phrases; variable-coefficient fractal pseudodifferential operators; and later citations. It found the source, its published version, [2], and an author research statement still presenting the assertion as a conjecture, but no later resolution. Novelty is therefore plausible but provisional.

Human-review recommendation. Review as a candidate full solution. The most sensitive point is the uniform dependence of the source’s constant-coefficient kernel estimates on a bounded family of S^0 seminorms. This dependence is explicit in that proof, and the source already uses it for the size estimate in its own variable-coefficient expansion.

References

- [1] M. Ionescu, L. G. Rogers, and R. S. Strichartz, *Pseudo-differential Operators on Fractals*, arXiv:1108.2246; Rev. Mat. Iberoam. 29 (2013), 1159–1190.
- [2] M. Ionescu and L. G. Rogers, *Complex Powers of the Laplacian on Affine Nested Fractals as Calderón–Zygmund Operators*, arXiv:1002.2011.